MIL-H-10201F
6 January 1986
SUPERSEDING
MIL-H-10201E
17 December 1969

MILITARY SPECIFICATION

HORNS, SOUND-WARNING, AIR-ACTUATED; HORNS, ELECTRICALLY-ACTUATED; AND BUZZERS, ELECTRICALLY-ACTUATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrically controlled, waterproof horns and buzzers used in 12 or 24 volt (V) direct current (dc) circuits as warning signals in tactical vehicles, combat vehicles, and other equipment (see 6.1).

1.2 Classification. Horns and buzzers covered by this specification shall be of the following types, classes, and categories, as specified (see 6.2 and 6.4):

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2590
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202640-17T13:40Z
Check the source to verify that this is the current version before use.

Type I - Horns.

Class 1 - Electrically actuated.
Category 1 - 12 V dc.
Category 2 - 24 V dc.

Class 2 - Air actuated.
Category 1 - 12 V dc.
Category 2 - 24 V dc.

Type II - Buzzers.

Class 1 - Electrically actuated.
Category 1 - 12 V dc.
Category 2 - 24 V dc.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

MILITARY

MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-P-52192	- Primer Coating, Epoxy.
MIL-P-53022	- Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free.
MIL-P-53030	- Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.

Source: https://assist.dla.mil -- Downloaded: 202640-17T13:40Z
Check the source to verify that this is the current version before use.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MIL-STD-45662	- Calibration Systems Requirements.
MS51074	- Horn, Electrical - Automotive Type, 12 or 24 Volts DC.
MS51301	- Horn, Sound-Warning, Air-Actuated, Electrically Controlled, 12 or 24 Volts DC.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

8675930	- Buzzer.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Horns and buzzers furnished under this specification shall be products which have been tested and have passed the qualification test specified herein, and have been listed in or approved for listing on the applicable Qualified Products List (QPL) (see 4.4 and 6.3).

Source: https://assist.dla.mil -- Downloaded: 202640-17T13:40Z
Check the source to verify that this is the currentversion before use.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Materials shall be free of defects which adversely affect performance or serviceability of the finished product (see 4.8.1 and 6.5).

3.3 Design and construction. Horns and buzzers shall be designed and constructed in accordance with the applicable drawings or standards listed in table I (see 4.5.2 and 4.8.2).

TABLE I. Applicable drawings or standards.

Type	Class	Category	MS part number or Army drawing number
I	1	1	MS51074-1
I	1	2	MS51074-2
I	2	1	MS51301-1
I	2	2	MS51301-2
II	1	1	8675930
II	1	2	8675930

3.4 Performance.

3.4.1 Sound performance.

3.4.1.1 Type I, class 1, and type II. When supplied with the nominal voltage input specified in table II, horns and buzzers shall meet the sound intensity and current requirements of table II (see 4.8.3.1).

3.4.1.2 Type I, class 2. When supplied with the nominal voltage and air pressure inputs specified in table II, horns shall meet the sound intensity, current, and air flow requirements of table II (see 4.8.3.1).

TABLE II. Sound performance requirements.

Type	Class	Category	Nominal voltage (V dc)	Reduced voltage (V dc)	Air pressure (psi)	Minimum sound intensity level [db(A)]	Maximum direct current (A)1/	Maximum air flow (cfm)2/
I	1	1	12	9		116	6	
I	1	2	24	18		116	3	
I	2	1	12	9	90	125	6	8.6
I	2	2	24	18	90	125	3	8.6
II	1	1	12	9		90	6	
II	1	2	24	18		90	3	

1/ Amperes (A)
2/ Cubic feet per minute (cfm)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:40Z
Check the source to verify that this is the current version before use.

3.4.2 Reduced voltage. Horns and buzzers shall produce sound within 5 decibels (A Weighted System Response) [db(A)] of the intensity level specified in table II while being operated at the reduced voltage specified in table II (see 4.8.3.2).

3.4.3 Reduced operating air pressure.

3.4.3.1 Type I, class 2 only. Horns shall produce sound within 5 db(A) of the intensity level specified in table II while being operated at reduced air pressures down to 60 pounds per square inch (psi) (see 4.8.3.3.1).

3.4.4 Sound quality.

3.4.4.1 Type I. Horns shall have a fundamental frequency between 100 and 500 cycles per second (cps) and the major noise-producing harmonic shall be not greater than 3500 cps when tested in accordance with 4.8.3.4.

3.4.4.2 Type II. Buzzers shall have a fundamental frequency between 105 and 185 cps when tested in accordance with 4.8.3.4.

3.4.5 Operating cycle.

3.4.5.1 Continuous operation. Horns and buzzers shall meet the requirements of 3.4.1 specified below and the requirements of 3.4.4 after being operated continuously for 15 minutes (see 4.8.3.5.1).

3.4.5.1.1 Type I. Horns shall meet the requirements of 3.4.1 within 5 db(A) of the intensity level specified in table II.

3.4.5.1.2 Type II. Buzzers shall evidence no deviation from the intensity level specified in table II.

3.4.5.2 Intermittent operation. Horns and buzzers shall meet the requirements of 3.4.1 specified below and the requirements of 3.4.4 after being operated for 100 000 cycles. One cycle of operation shall consist of 1 second on (energized) and 2 seconds off (see 4.8.3.5.2).

3.4.5.2.1 Type I. See 3.4.5.1.1.

3.4.5.2.2 Type II. See 3.4.5.1.2.

3.4.6 Insulation resistance. The electrical insulation shall evidence no cracking, charring, loosening, burning, smoking, or other damage after the horn or buzzer is tested in accordance with 4.8.3.6.

3.4.7 Connector and terminal strength. Horns and buzzers shall meet the requirements of 3.4.8.3 after the connectors and terminals are subjected to a 50-pound pull (see 4.8.3.7).

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:40Z
Check the source to verify that this is the current version before use.

3.4.8 Environmental.

3.4.8.1 High temperature. After being stabilized at 160 degrees Fahrenheit (°F), horns and buzzers shall produce sound within 5 db(A) of the intensity level specified in table II (see 4.8.3.8.1).

3.4.8.2 Low temperature. After being stabilized at -65°F, horns and buzzers shall produce sound within 5 db(A) of the intensity level specified in table II (see 4.8.3.8.2).

3.4.8.3 Water immersion. Horns and buzzers shall evidence no air bubbles escaping from any sealed chamber when submerged in a salt water solution and shall subsequently meet the requirements of 3.4.1 and 3.4.4 (see 4.8.3.8.3).

3.4.8.3.1 Valve airtightness.

3.4.8.3.1.1 Type I, class 2 only. Horns shall evidence no bubbles escaping past the air valve from the air valve inclosure after being connected to a 90 psi air source and submerged in water (see 4.8.3.8.3.1.1).

3.4.8.4 Salt fog. Horns and buzzers shall meet the requirements of 3.4.1 and 3.4.4 and shall evidence no corrosion after exposure to a salt fog atmosphere (see 4.8.3.8.4).

3.4.8.5 Fungus. Horns and buzzers shall show no evidence of fungus growth affecting operation and shall meet the requirements of 3.4.1 and 3.4.4 after exposure to conditions favorable to fungus growth (see 4.8.3.8.5).

3.4.8.6 Vibration. Horns and buzzers shall show no evidence of disintegration of component materials or assemblies and shall meet the requirements of 3.4.1 and 3.4.4 after being subjected to vibration in each direction of the three major axes. The amplitude and frequency shall be as specified in MIL-STD-202, method 201 (see 4.8.3.8.6).

3.4.8.7 Shock. Horns and buzzers shall meet the requirements of 3.4.1 and 3.4.4 after being subjected to induced shocks of sawtooth waveform along each of the three mutually perpendicular axes with a peak value of 100 gravity units (g) for 6 milliseconds (ms) (see 4.8.3.8.7).

3.4.8.8 Electromagnetic interference. Horns and buzzers shall meet the requirements of MIL-STD-461 for class A3 vehicle components, methods CE04 and RE02 (see 4.8.3.8.8).

3.5 Finish. Unless otherwise specified (see 6.2), the outer surface of the horn or buzzer shall be cleaned in accordance with the applicable method of TT-C-490. The cleaned surfaces shall then be treated with a coating conforming to type I of TT-C-490 after which the surfaces shall be primed with primer conforming to MIL-P-52192, MIL-P-53030, or MIL-P-53022, to a thickness of 0.08 to 1.2 mils. Unless otherwise specified (see 6.2), the outer surfaces shall then be coated with a chemical agent resistant coating conforming to color green 383 per MIL-C-46168 or MIL-C-53039, to a thickness of at least 1.8 mils (see 4.5.2 and 4.8.2).

Source: https://assist.dla.mil -- Downloaded: 20260-17T13:40Z
Check the source to verify that this is the current version before use.

3.6 Identification and markings. The following information shall be permanently and legibly marked on each horn and buzzer, as applicable, in accordance with MIL-STD-130 (see 4.5.2 and 4.8.2):

Horn, Air-Operated (Automotive) or, Horn, Electrical (Automotive), or Buzzer, Electrical.
Pressure rating, when applicable.
Military or MS part number.
Manufacturer's part or serial number.
Manufacturer's name or trademark.
Other information deemed necessary by the manufacturer or required on the applicable drawing.

3.7 Workmanship. Workmanship shall be such quality as to assure that horns and buzzers furnished under this specification are free of defects that compromise, limit, or reduce performance or intended use. Horns and buzzers shall be free of burrs, scratches, rust, chips, projections, and sharp edges or any condition that might present a safety hazard to operating or maintenance personnel (see 4.5.2 and 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply wih all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

Source: https://assist.dla.mil -- Downloaded: 202640-17T13:40Z
Check the source to verify that this is the currenversion before use.

4.2 Classification of inspection:

a. Qualification inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 ± 18°F
b. Barometric pressure 28.5 + 2 inches mercury (Hg)
   - 3
c. Relative humidity 50 ± 30 percent

4.4 Qualification inspection. A qualification sample of four horns or buzzers of the specified type, class, and category shall be utilized for qualification testing. The horns or buzzers shall be representative of the units proposed to be furnished under the contract. Qualification testing shall be conducted under Government surveillance by the contractor, or an authorized testing facility, at a site approved by the Government. Inspection shall consist of testing as specified in table III, in the order listed in table IV. Qualification inspection shall be performed at a laboratory acceptable to the Government on sample units produced with equipment and procedures normally used in inspection (see 6.3).

4.4.1 Retention of qualification. Certification shall be requested every two years from each manufacturer listed on the QPL to retain listing on the QPL. This certification shall be forwarded to the preparing activity and shall be signed by a responsible official of management attesting that the listed product still meets the requirements of the current issue of the specification, is available from the listed plant, and can be produced under the same conditions as originally qualified; that is, same process, materials, construction, design, manufacturer's part number or designation. Failure to provide certification shall be cause for removal from the QPL.

4.4.2 Failure. Failure of a qualification sample to pass any of the inspections specified herein may be cause for the Government to refuse to conduct additional inspections until the faults revealed by the inspection have been corrected.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the horns or buzzers of one type, class, category, and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 202640-17T13:40Z
Check the source to verify that this is the current version before use.

4.5.1.3 Sampling for test. Samples for tests shall be selected in accordance with level S-3 of MIL-STD-105.

TABLE III. Classification of inspections.

Title	Requirement	Inspection	Qualification samples	Quality conformance		
				Examination	Test	Control
Materials	3.2	4.8.1	X			
Defects (see 4.5.2 and table V)	3.3, 3.5, 3.6, and 3.7	4.8.2	X	X		
Sound intensity	3.4.1	4.8.3.1	X		X	
Reduced voltage	3.4.2	4.8.3.2	X			
Reduced operating air pressure	3.4.3	4.8.3.3	X			
Sound performance	3.4.4	4.8.3.4	X		X	
Continuous operation	3.4.5.1	4.8.3.5.1	X			X
Intermittent operation	3.4.5.2	4.8.3.5.2	X			
Insulation resistance	3.4.6	4.8.3.6	X			X
Connector and terminal strength	3.4.7	4.8.3.7	X			
High temperature	3.4.8.1	4.8.3.8.1	X			
Low temperature	3.4.8.2	4.8.3.8.2	X			
Water immersion	3.4.8.3	4.8.3.8.3	X			
Valve air-tightness	3.4.8.3.1	4.8.3.8.3.1	X			
Salt fog	3.4.8.4	4.8.3.8.4	X			
Fungus	3.4.8.5	4.8.3.8.5	X			
Vibration	3.4.8.6	4.8.3.8.6	X			
Shock	3.4.8.7	4.8.3.8.7	X			
Electromagnetic interference	3.4.8.8	4.8.3.8.8	X			

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:40Z
Check the source to verify that this is the current version before use.

TABLE IV. Qualification test sequence.

Sample number			
1	2	3	4
4.8.1	4.8.1	4.8.1	4.8.1
4.8.2	4.8.2	4.8.2	4.8.2
4.8.3.5.1	4.8.3.1	4.8.3.8.8	4.8.3.1
4.8.3.1	4.8.3.8.1	4.8.3.1	4.8.3.8.1
4.8.3.8.1	4.8.3.8.2	4.8.3.8.1	4.8.3.8.2
4.8.3.8.2	4.8.3.3	4.8.3.8.2	4.8.3.3
4.8.3.3	4.8.3.2	4.8.3.3	4.8.3.2
4.8.3.2	4.8.3.4	4.8.3.2	4.8.3.4
4.8.3.4	4.8.3.8.3.1 1/	4.8.3.4	4.8.3.8.3.1 1/
4.8.3.8.3.1 1/	4.8.3.5.2	4.8.3.8.3.1 1/	4.8.3.8.7
	4.8.3.1	4.8.3.8.6	4.8.3.1
	4.8.3.4	4.8.3.1	4.8.3.4
	4.8.3.8.3.1 1/	4.8.3.4	4.8.3.8.3.1 1/
	4.8.3.7	4.8.3.8.3.1 1/	4.8.3.8.4
	4.8.3.8.3	4.8.3.8.5	4.8.3.1
	4.8.3.1	4.8.3.1	4.8.3.4
	4.8.3.4	4.8.3.4	
	4.8.3.8.3.1 1/		
	4.8.3.6		

1/ Type I, class 2 only.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table V.

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:40Z
Check the source to verify that this is the current version before use.

TABLE V. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3).	SIE 1/
103	Identification marking, improper (see 3.6).	Visual
104	Faulty workmanship affecting performance (see 3.7).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3).	SIE 1/
202	Finish, improper (see 3.5).	Visual
203	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Tests. Samples selected in accordance with 4.5.1.3 shall be subjected to the quality conformance tests specified in table III, and shall conform to an AQL of 6.5 based on percent defective.

4.6 Control tests. Control tests shall be conducted on two horns or buzzers from each lot of 500 units consecutively produced, except that not more than four nor less than two tests shall be performed in a 30-day period. The horns and buzzers shall be subjected to the control tests specified in table III, in the order listed.

4.7 Failure. Failure of any horn or buzzer to pass any of the specified quality conformance or control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

Source: https://assist.dla.mil -- Downloaded: 202640-17T13:40Z
Check the source to verify that this is the current version before use.

4.8.2 Defects. Conformance to 3.3, 3.5, 3.6, and 3.7 shall be determined by examination for the defects listed in table V. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Performance.

4.8.3.1 Sound performance. To determine conformance to 3.4.1, horns and buzzers shall be tested as specified below.

4.8.3.1.1 Apparatus. The test apparatus shall include the following:

a. A power source consisting of an automotive battery of the correct nominal voltage as specified in table II, equipped with an electrical power supply connected in parallel to maintain the state of a constant fully charged battery. The power supply's terminal voltage shall not contain a ripple component of more than 1 V peak to peak, as measured by an oscilloscope, when operating the test specimen.
b. An air source with means for controlling and measuring air pressure and airflow (volume).
c. A sound level meter.
d. A sound analyzer.
e. A voltmeter.
f. An ammeter.
g. A sound level calibrator.

4.8.3.1.2 Procedure.

4.8.3.1.2.1 Type I, class 1. Mount the horn so that the sound waves which are emitted from the horn will be directed towards the sound pickup on the sound level meter. The horn shall be at least 4 feet from the floor and 10 feet from any wall surface so as to minimize sound reflection. The distance between the horn and sound pickup shall be varied to obtain the greatest meter reading possible within the range of 5 $\pm$ 0.5 feet. Ambient temperature shall be maintained at 77 $\pm$ 15°F. The horn shall be operated at the nominal voltage specified in table II. Sound intensity and current consumption shall be measured and recorded to determine conformance to 3.4.1.

4.8.3.1.2.2 Type I, class 2. Horns shall be tested as specified in 4.8.3.1.2.1 while operating at the nominal voltage and air pressure specified in table II. Sound intensity, current consumption, and airflow shall be measured and recorded to determine conformance to 3.4.1.

4.8.3.1.2.3 Type II. Connect the buzzer to the power source. The buzzer shall be suspended freely on a wire with the pickup microphone positioned 4 inches from the face of, and on the centerline with, the buzzer. The buzzer shall be operated at the nominal voltage specified in table II. The sound intensity and current consumption shall be measured and recorded to determine conformance to 3.4.1.

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:40Z
Check the source to verify that this is the current version before use.

4.8.3.2 Reduced voltage. To determine conformance to 3.4.2, horns and buzzers shall be tested as specified in 4.8.3.1 except that the horns and buzzers shall be operated at the reduced voltages specified in table II. The sound intensity shall be measured and recorded.

4.8.3.3 Reduced operating air pressure.

4.8.3.3.1 Type I, class 2 only. To determine conformance to 3.4.3.1, horns shall be tested as specified in 4.8.3.1 except that the horns shall be operated by air pressures of 60, 70, and 80 psi. The sound intensity shall be measured and recorded at each air pressure.

4.8.3.4 Sound quality. To determine conformance to 3.4.4, horns and buzzers shall be tested as specified in 4.8.3.1, as applicable, except that the sound level meter shall be connected to a sound analyzer. The frequency of the fundamental and of the most intense harmonic shall be measured and recorded.

4.8.3.5 Operating cycle.

4.8.3.5.1 Continuous operation. To determine conformance to 3.4.5.1, an untested horn or buzzer shall be mounted as in intended use and shall be continuously operated for 15 minutes. At the end of the 15-minute period, the horn or buzzer shall be allowed to return to ambient air temperature and shall be tested as specified in 4.8.3.1 and 4.8.3.4.

4.8.3.5.2 Intermittent operation. To determine conformance to 3.4.5.2, horns and buzzers shall be mounted as in intended use and operated for 100 000 cycles. At the completion of 100 000 cycles, the horns and buzzers shall be allowed to return to ambient air temperature and shall be tested as specified in 4.8.3.1 and 4.8.3.4.

4.8.3.5.3 Type I, class 2 only. Before and after testing as specified in 4.8.3.5.1 and 4.8.3.5.2, horns shall meet the requirements of 3.4.8.3.1.1.

4.8.3.6 Insulation resistance. To determine conformance to 3.4.6, horns and buzzers shall be subjected to 500 V root mean square (rms) at a frequency of 60 cps, applied 1 minute between each terminal and ground, with the internal grounds disconnected. The horns and buzzers shall then be visually examined.

4.8.3.7 Connector and terminal strength. To determine conformance to 3.4.7, the connectors and terminals of the horns and buzzers shall be subjected to a 50-pound pull for 15 minutes, first in a direction normal to and then in a direction parallel to their respective longitudinal axes. The normal axes pull shall be made from their mating ends. The horns and buzzers shall then be tested as specified in 4.8.3.8.3.

4.8.3.8 Environmental.

4.8.3.8.1 High temperature. To determine conformance to 3.4.8.1, horns and buzzers shall be tested as specified in 4.8.3.1 except that the temperature of the air and the horn or buzzer shall be stabilized at 160°F.

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:40Z
Check the source to verify that this is the currentversion before use.

4.8.3.8.2 Low temperature. To determine conformance to 3.4.8.2, horns and buzzers shall be tested as specified in 4.8.3.1 except that the temperature of the air and the horn or buzzer shall be stabilized at -65°F.

4.8.3.8.3 Water immersion. To determine conformance to 3.4.8.3, horns and buzzers shall be tested in accordance with MIL-STD-1184, type II, class 1, method 100, procedure I.

4.8.3.8.3.1 Valve airtightness.

4.8.3.8.3.1.1 Type I, class 2 only. To determine conformance to 3.4.8.3.1.1, horns shall be connected to a 90 psi air source, with the control disconnected. The horns shall be submerged in water for 10 minutes with the projector opening towards the air-water boundary. There shall be no evidence of air bubbles.

4.8.3.8.4 Salt fog. To determine conformance to 3.4.8.4, horns and buzzers shall be tested in accordance with MIL-STD-202, method 101. A 5-percent salt solution shall be used and the duration of the test shall be 200 hours. After testing, horns and buzzers shall be subjected to the tests specified in 4.8.3.1 and 4.8.3.4.

4.8.3.8.5 Fungus. To determine conformance to 3.4.8.5, horns and buzzers shall be tested in accordance with MIL-F-13927, class II, method A, except that the performance test shall be conducted at the end of 90 days of incubation only. Horns and buzzers shall be tested for performance as specified in 4.8.3.1 and 4.8.3.4.

4.8.3.8.6 Vibration. To determine conformance to 3.4.8.6, horns and buzzers shall be tested in accordance with MIL-STD-202, method 201. Motion shall be applied for at least 1 hour in the direction of each of the three major axes. After testing, horns and buzzers shall be subjected to the tests specified in 4.8.3.1 and 4.8.3.4.

4.8.3.8.6.1 Type I, class 2 only. Before and after testing as specified in 4.8.3.8.6, horns shall meet the requirements of 3.4.8.3.1.1.

4.8.3.8.7 Shock. To determine conformance to 3.4.8.7, horns and buzzers shall be mounted as in intended operation and shall be tested in accordance with MIL-STD-202, method 213B, condition I. Two induced shocks of sawtooth waveform shall be applied along each of the three mutually perpendicular axes. After testing, horns and buzzers shall be subjected to the tests specified in 4.8.3.1 and 4.8.3.4.

4.8.3.8.7.1 Type I, class 2 only. Before and after testing as specified in 4.8.3.8.7, horns shall meet the requirements of 3.4.8.3.1.1.

4.8.3.8.8 Electromagnetic interference. To determine conformance to 3.4.8.8, horns and buzzers shall be tested in accordance with MIL-STD-462, class A3 vehicle components, methods CE04 and RE02.

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:40Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. Horns and buzzers covered by this specification are intended for use as warning signals on tactical vehicles, combat vehicles, or other equipment operating under severe and extreme conditions of shock, vibration, and climate.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, category, and drawing or MS standard title, number, and date (see 1.2 and 3.3).
c. If finish is to be other than as specified (see 3.5).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
f. If inspection conditions shall be other than as specified (see 4.3).
g. Selection of applicable levels of preservation, packaging, packing, and marking (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion on the applicable QPL, whether or not such products have actually been so listed by that date. The attention of contractors is called to this requirement, and manufacturers are urged to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the parts covered by this specification. The activity responsible for the QPL is the Commanding General, US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000 and information pertaining to qualification of products may be obtained from that activity (see 3.1 and 4.4).

6.4 Cross-reference. Table VI provides a cross-reference for classification between this specification and the previous revision (see 1.2).

Source: https://assist.dla.mil -- Downloaded: 202640-17T13:40Z
Check the source to verify that this is the current version before use.

TABLE VI. Classification cross-reference.

Former classification designation	New classification designation
Type I	Type I, class 2
Style 1	Deleted
Style 2	Deleted
Type II	Type I, class 1
Type III	Type II, class 1
Type IV	Deleted

6.5 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.6 Changes from previous issue. Asterisks, to identify changes with respect to the previous issue, are not used in this revision due to the extensiveness of the changes.

Custodian:
Army - AT
Air Force - 99

Preparing activity:
Army - AT

(Project No. 2590-0202)

Review activities:
Army - ER
DLA - CS
Air Force - 84

User activities:
Army - ME
Navy - MC

Source: https://assist.dla.mil -- Downloaded: 202640-17T13:40Z
Check the source to verify that this is the currenversion before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Tank-Automotive Command
ATTN: DRSTA-GSS
Warren, MI 48080

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 20260-17113:40Z
Check the source to verify that this is the current version before use.